INDUSTRIAL VISIT REPORT – KIRUTHIGA INDUSTRIES

1. Introduction

As part of our academic curriculum, we visited **Kiruthiga Industries** to gain practical knowledge about modern manufacturing and machining processes. The visit provided us with an opportunity to observe the practical application of concepts studied in Mechanical Engineering, particularly CNC machining, manufacturing processes, metrology, and quality control.

During the visit, we learned about the manufacturing and machining of **motor-pump components**, including components used for connecting pumps and motors. We were introduced to CNC turning, milling, Vertical Machining Centres (VMC), CNC programming, tooling, dimensional tolerances, and inspection methods.

2. Objectives of the Industrial Visit

The main objectives of the visit were:

- To understand practical manufacturing processes used in industries.
- To observe CNC turning and milling operations.
- To understand the working of advanced CNC machines.
- To learn the basics of CNC programming and machine control.
- To understand dimensional tolerance and precision machining.
- To observe measurement and inspection techniques.
- To understand the importance of quality control in manufacturing.

3. Manufacturing and Machining Processes

One of the major areas discussed during the visit was the machining of **motor-pump components**. The pump and motor are connected using suitable components such as adaptors, which require accurate machining for proper alignment and operation.

The manufacturing process involves machining the required component to the specified dimensions followed by inspection. Proper dimensional accuracy is important because errors in the component can affect assembly, alignment, and the performance of rotating machinery.

We observed that CNC machines are widely used for such manufacturing operations because they provide high accuracy, repeatability, and productivity.

4. CNC Turning and Milling

We were introduced to **CNC turning**, which is mainly used for manufacturing cylindrical and rotational components. In CNC turning, the workpiece rotates while the cutting tool removes material according to a programmed tool path. Operations such as facing, turning, grooving, boring, and threading can be performed.

We also learned about **CNC milling and Vertical Machining Centres (VMCs)**. Milling uses a rotating cutting tool to remove material from the workpiece. The notes taken during the visit included **3-axis milling and 4/5-axis VMC machining**. Multi-axis machines provide greater flexibility and are useful for machining complex components and surfaces.

The use of advanced CNC machines improves production accuracy and reduces the variation between components.

5. CNC Control and Programming

During the visit, we were introduced to CNC control systems such as **FANUC and Siemens**. The CNC controller interprets the machining program and controls the movement of the machine, spindle, feed, and tools.

We also learned about CNC programming involving the **X and Z axes**, particularly in turning operations. CNC programs contain instructions that control tool movement, spindle operation, feed rate, and machining sequences.

The machine can also accommodate multiple tools through different tool stations. The notes from the visit mention an **8-station tool arrangement**, which allows different machining operations to be performed efficiently without frequent manual tool changes.

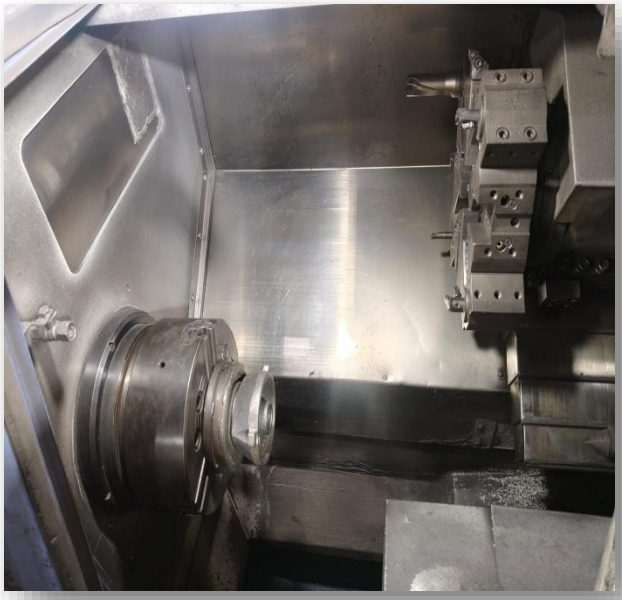


Fig. 1 CNC chuck and tools



Fig. 2 CNC control & program

6. Dimensional Tolerance and Inspection

An important concept observed during the visit was **dimensional tolerance**. In manufacturing, it is practically difficult to produce every component at exactly its nominal dimension. Therefore, a permissible range of variation is provided.

The notes taken during the visit mention tolerances of approximately **0.02 mm and 0.1 mm**, depending on the component and machining requirement. Such small tolerances demonstrate the importance of precision machining in industrial component manufacturing.

We also learned about the importance of measuring and inspection equipment. Instruments such as **Vernier measuring instruments and dial gauges** can be used to check dimensions, alignment, and small deviations. Precision inspection ensures that the manufactured components satisfy the required specifications before assembly or further processing.

7. Materials and Quality Control

The visit also included discussions related to **iron-based/carbon materials** and components associated with industrial applications such as valves and rotating machinery. The selection of material depends on the required strength, durability, wear resistance, and operating conditions.

Quality control is an essential part of manufacturing. After machining, components must be inspected to ensure that their dimensions are within the specified tolerance. Proper inspection helps reduce manufacturing defects, assembly problems, material wastage, and machine downtime.

The visit helped us understand the relationship between **machining, dimensional accuracy, measurement, and product quality**.

8. Learning Outcomes

The industrial visit provided us with valuable practical knowledge about modern manufacturing. We gained an understanding of:

- CNC turning and milling operations.
- VMC and multi-axis machining.
- CNC programming and machine control.
- FANUC and Siemens CNC control systems.
- Tool stations and machining tools.
- Dimensional tolerance and precision manufacturing.
- Use of measuring and inspection instruments.
- Manufacturing of motor-pump-related components.
- Importance of quality control and productivity.

The visit also helped us connect classroom concepts from **Manufacturing Processes, Engineering Metrology, CNC Machines, CAD/CAM, and Production Engineering** with their practical applications.

9. Conclusion

The industrial visit to **Kiruthiga Industries** was a valuable and informative experience for us as Mechanical Engineering students. It provided practical exposure to CNC machining, motor-pump component manufacturing, turning, milling, VMC operations, CNC programming, tooling, dimensional tolerances, and inspection techniques.

The visit helped us understand the importance of **accuracy, precision, repeatability, quality, and productivity** in modern manufacturing industries. It also gave us a better understanding of how theoretical concepts learned in the classroom are applied in real industrial environments.

Overall, the visit enhanced our technical knowledge and provided useful practical exposure to modern manufacturing technologies, which will be valuable for our future academic studies and careers in Mechanical Engineering.

